Experiment 12: Analysis of Advanced Optical Modulation Formats Using OptiSystem

OBJECTIVES

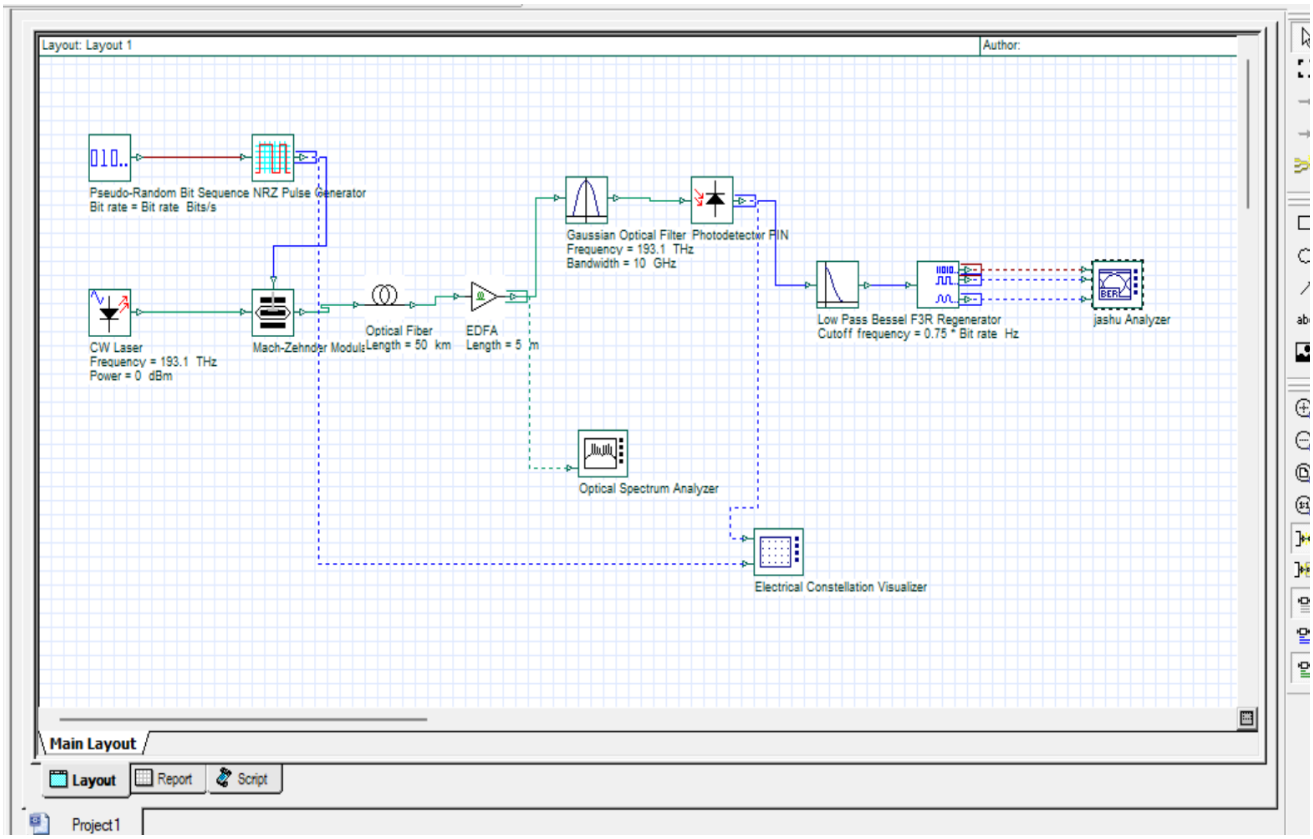
1. Design and simulate OOK, PAM-4, DPSK and QAM transmitter–receiver systems in OptiSystem.
2. Analyze the signal levels, constellation diagrams, eye diagrams and spectra of each modulation format.
3. Evaluate BER and Q-factor of an OOK link and observe the eye diagrams of the multilevel (M-ary) formats.
4. Compare the modulation formats in terms of bits per symbol, spectral efficiency and noise tolerance.

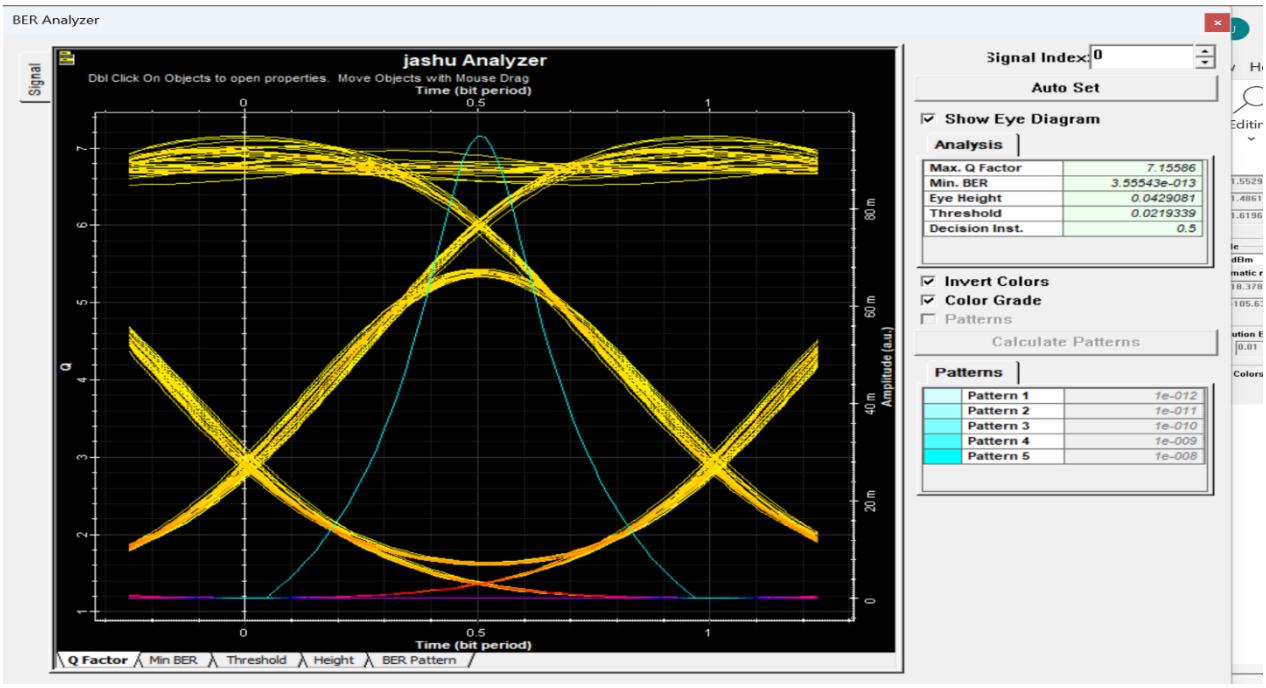
SOFTWARE REQUIRED

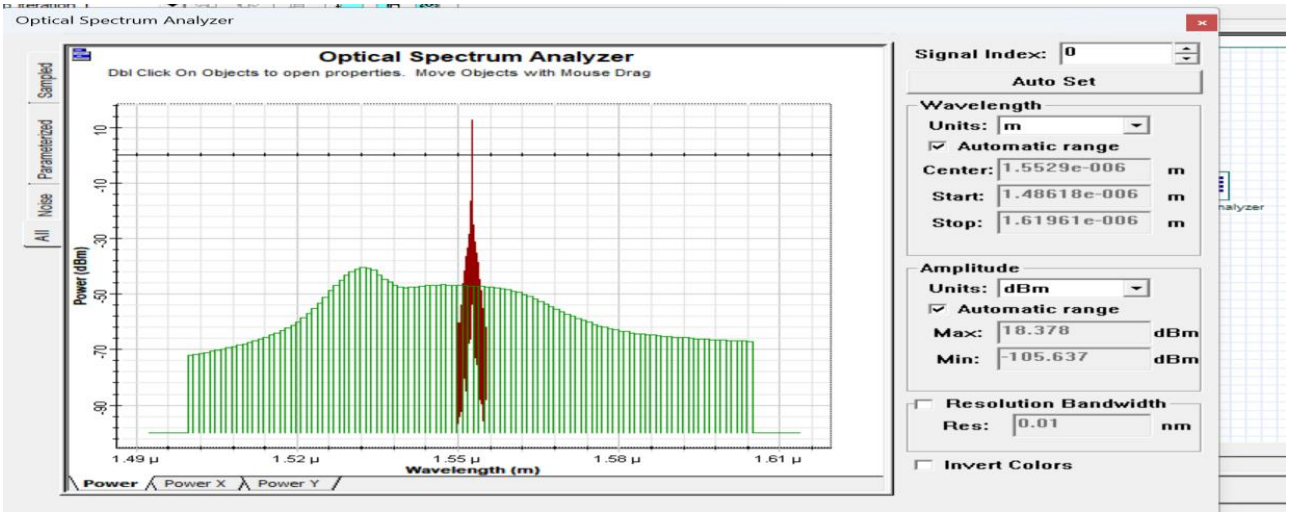
- Optisystem

BLOCK DIAGRAMS

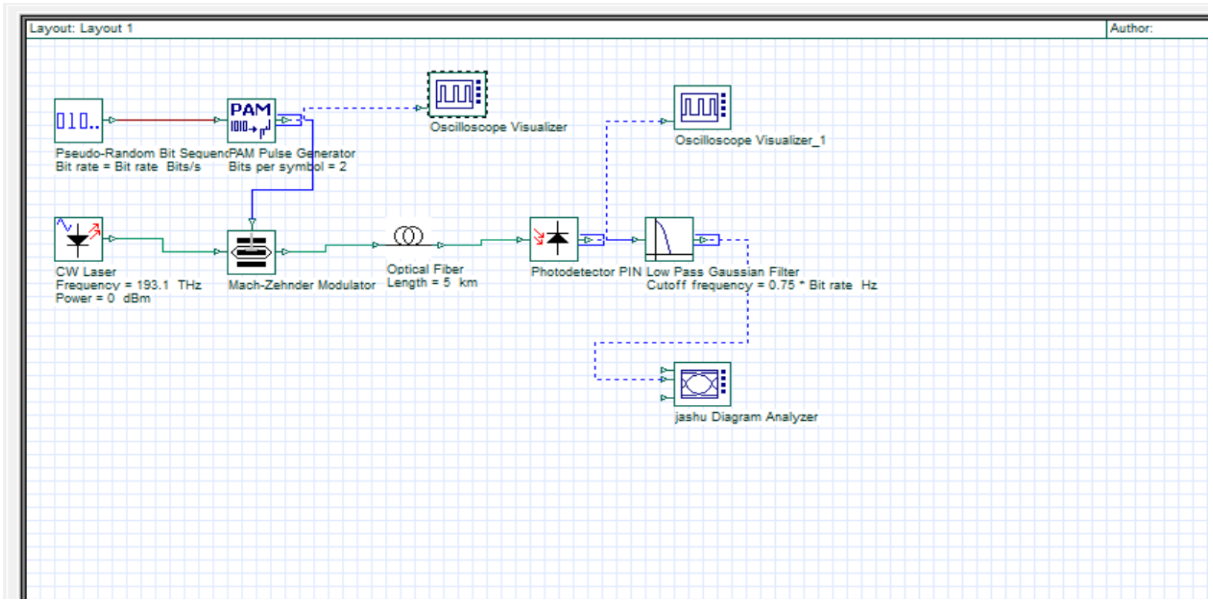
OOK transmitter and receiver

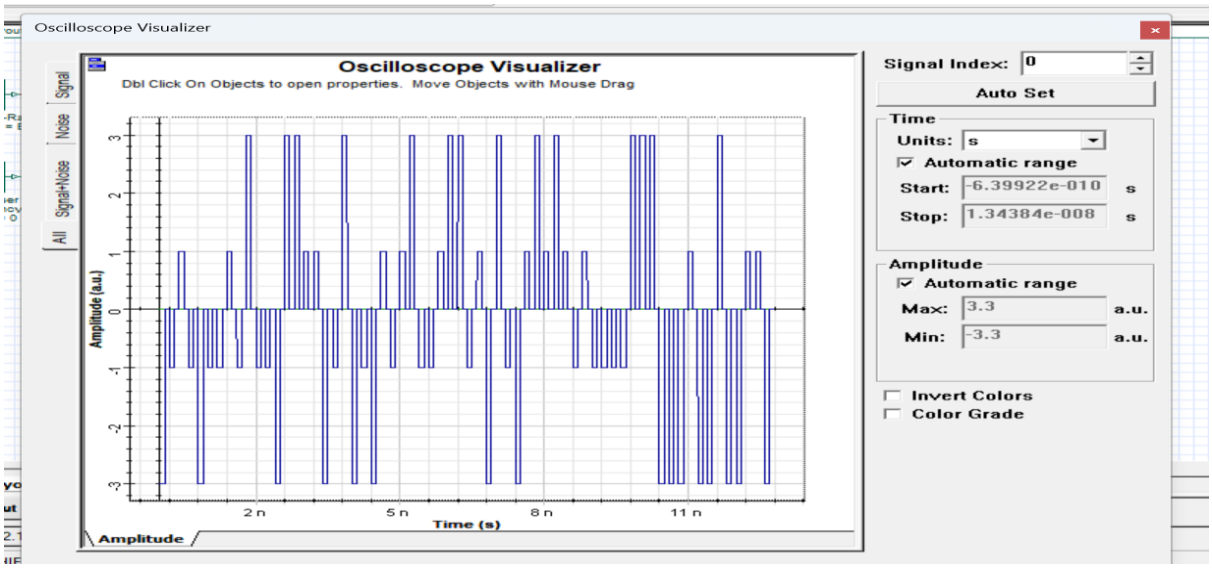


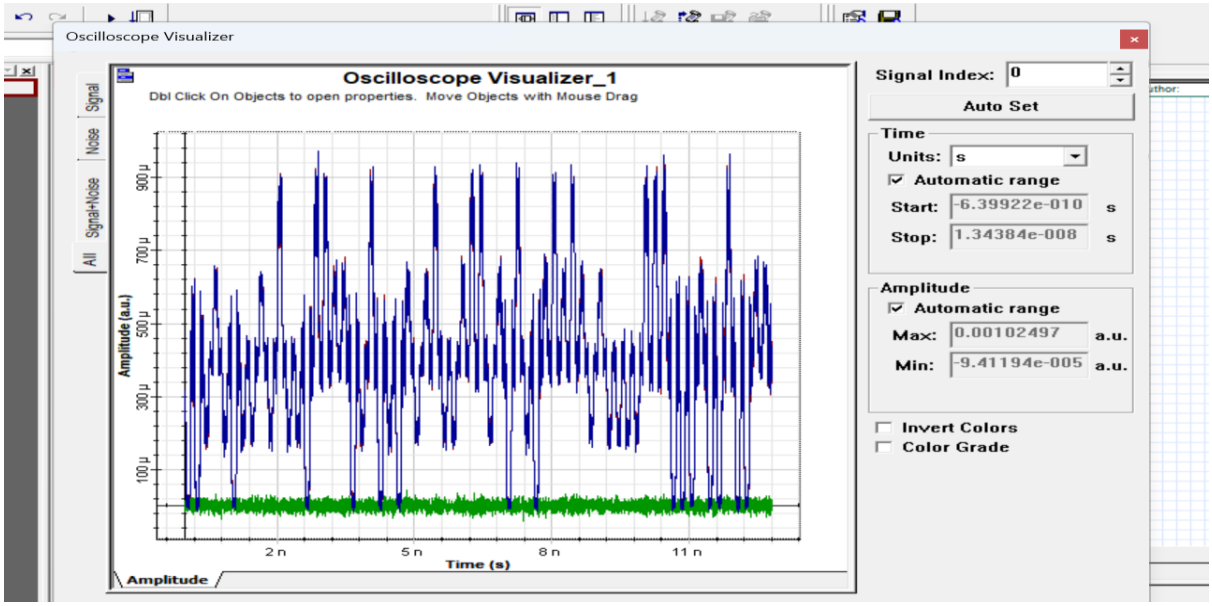




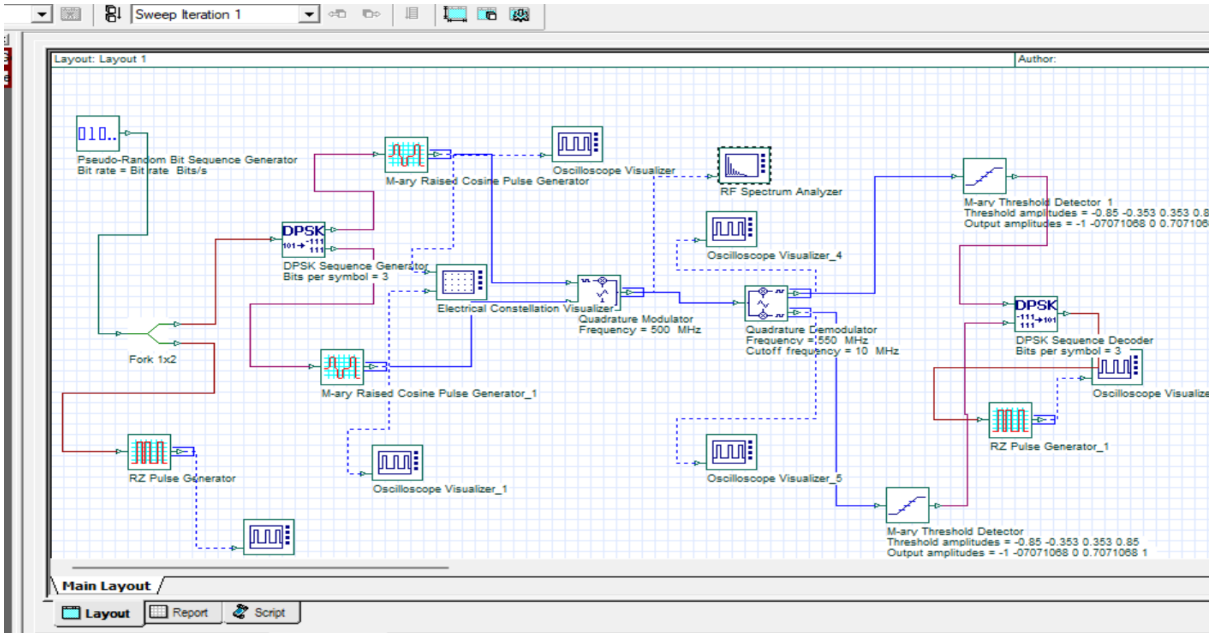
PAM-4 transmitter and receiver

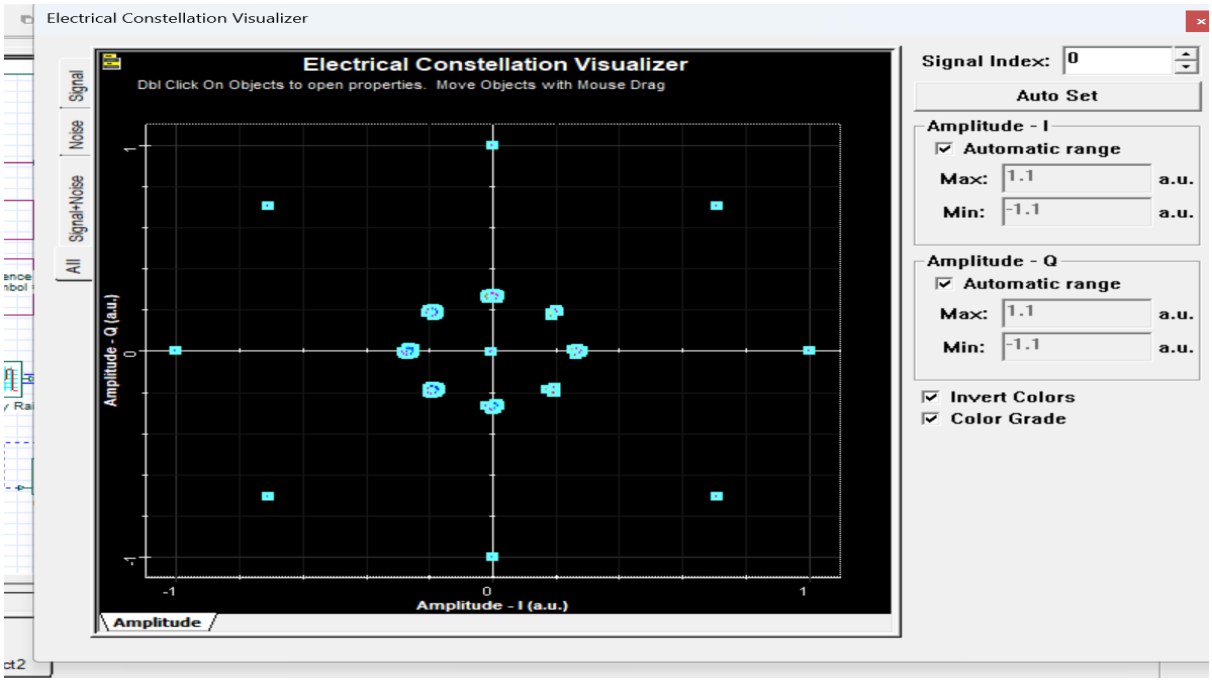


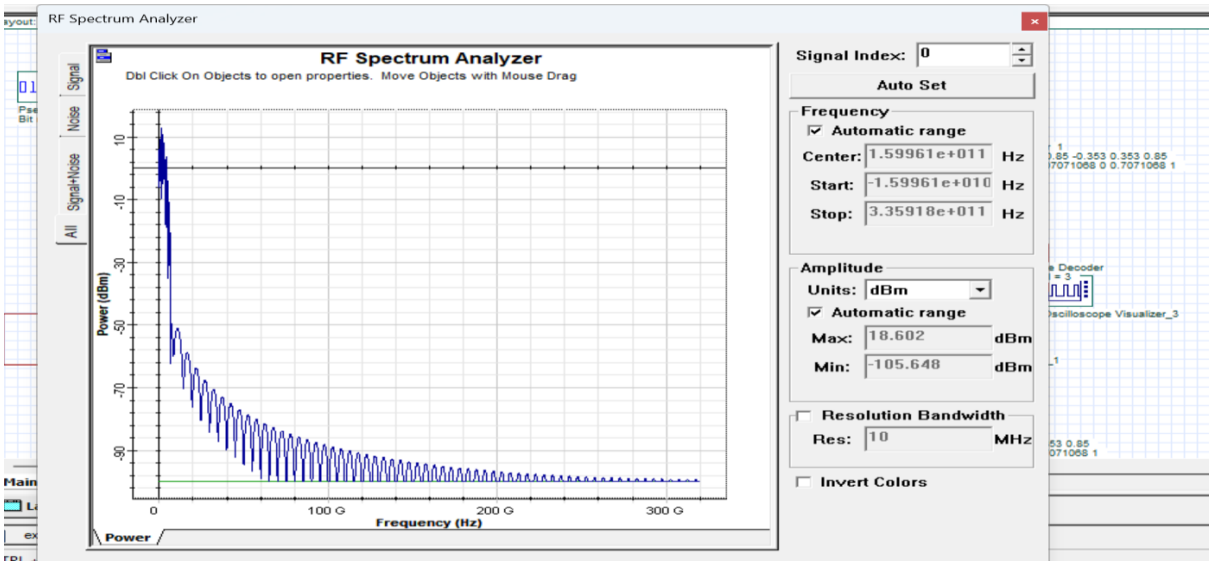


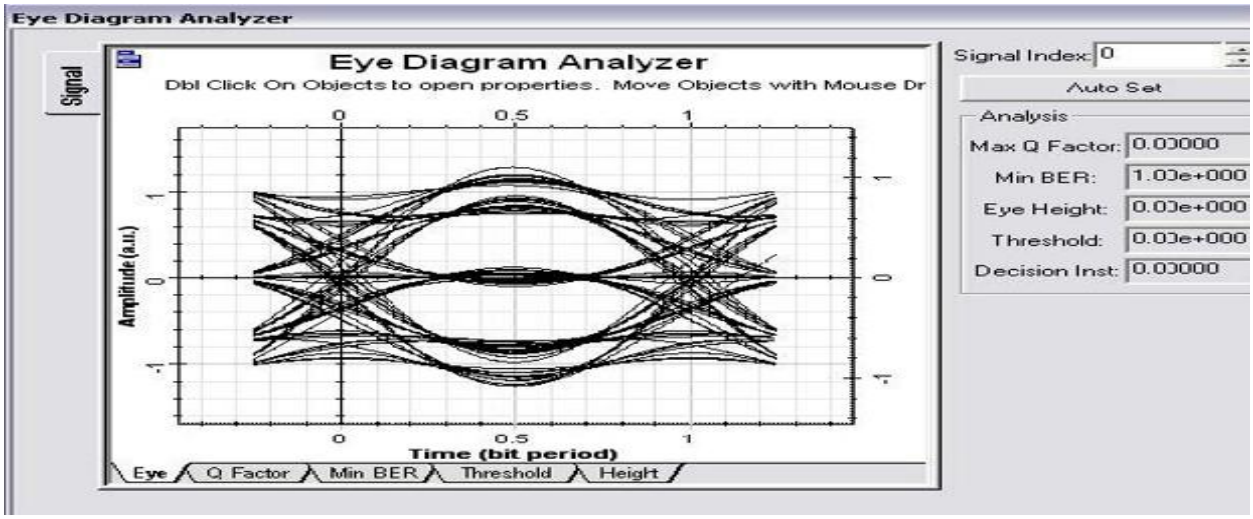


8-DPSK transmitter and receiver

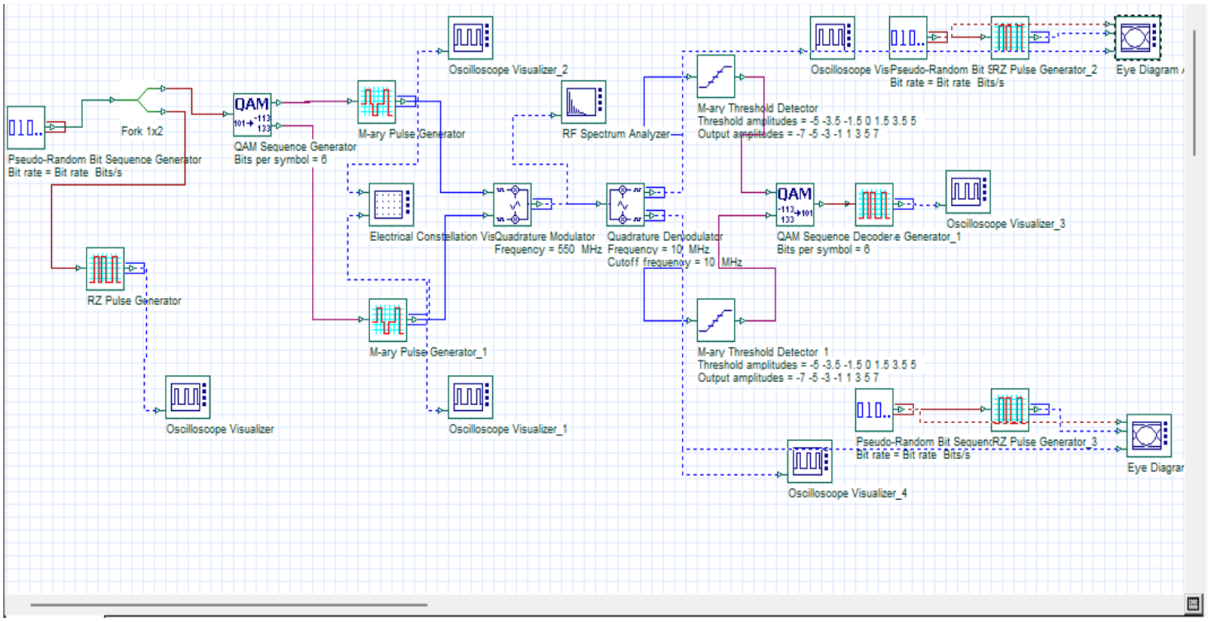


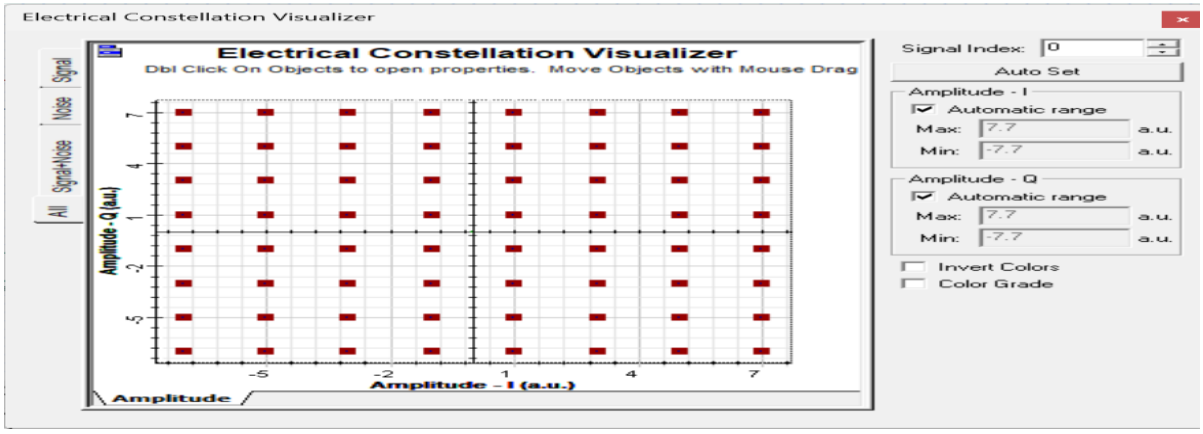

Electrical Constellation Visualizer
Electrical Constellation Visualizer
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Signal
Noise
Signal+Noise
All
Amplitude - Q (a.u.)
Amplitude - I (a.u.)
Amplitude
Signal Index: 0
Auto Set
Amplitude - I
Automatic range
Max: 1.1 a.u.
Min: -1.1 a.u.
Amplitude - Q
Automatic range
Max: 1.1 a.u.
Min: -1.1 a.u.
Invert Colors
Color Grade

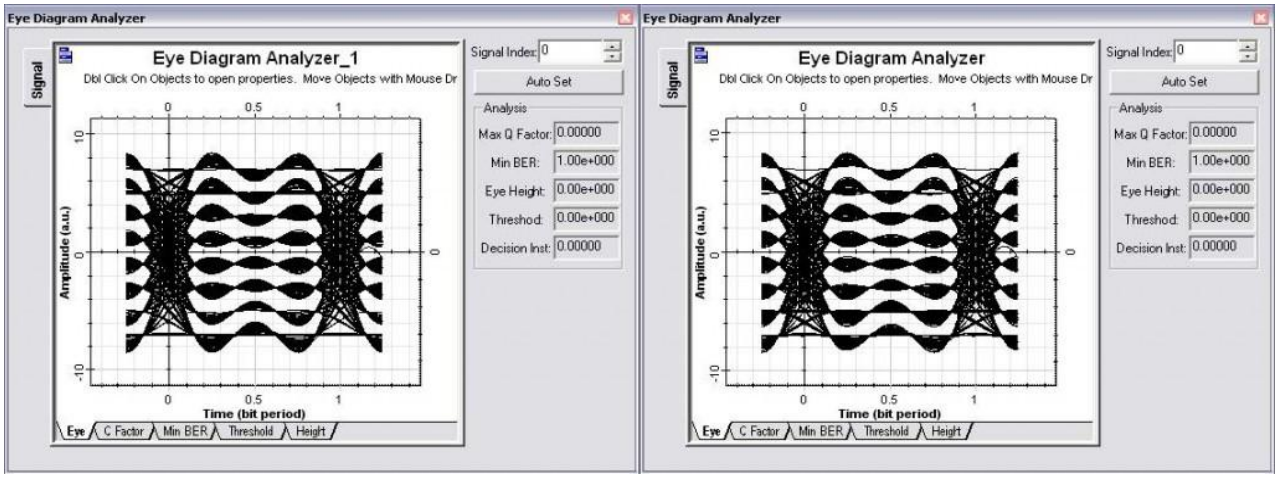

RF Spectrum Analyzer
RF Spectrum Analyzer
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Signal
Noise
Signal+Noise
All
Power (dBm)
Frequency (Hz)
Power
Signal Index: 0
Auto Set
Frequency
Automatic range
Center: 1.59961e+011 Hz
Start: -1.59961e+010 Hz
Stop: 3.35918e+011 Hz
Amplitude
Units: dBm
Automatic range
Max: 18.602 dBm
Min: -105.648 dBm
Resolution Bandwidth
Res: 10 MHz
Invert Colors


Eye Diagram Analyzer
Eye Diagram Analyzer
Dbl Click On Objects to open properties. Move Objects with Mouse Dr
Signal
Amplitude (a.u.)
Time (bit period)
Eye
Q Factor
Min BER
Threshold
Height
Signal Index: 0
Auto Set
Analysis
Max Q Factor: 0.00000
Min BER: 1.00e+000
Eye Height: 0.00e+000
Threshold: 0.00e+000
Decision Inst: 0.00000

64-QAM







REFLECTIONS:I learned about advanced optical modulation formats and their importance in high-speed communication. This experiment helped me understand how modulation techniques improve transmission capacity and efficiency.

LABORATORY PERFORMANCE ASSESSMENT RUBRIC

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
System	Correct selection and	All transmitter,	All	Component	Incorrect	20%
Design &	connection of	channel, receiver	components	selection is	component	
Modulation	OptiSystem	and visualizer	are present	partially	selection or	
Setup	components for each	blocks for the	with minor	complete;	improper	
	format (PRBS,	four formats are	connection or	connection or	setup leading	
	NRZ/PAM/DPSK/QAM	correctly	layout errors,	configuration	to simulation	
	generators, Mach-	selected,	but the	errors impede	failure.	
	Zehnder or Quadrature	connected in	systems	signal flow.		
	Modulator, fiber, PIN,	sequence and	remain			
	filters, threshold	configured	functional.			
	detectors, decoders,	without errors.				
	visualizers).					
Parameter	Accurate setup of bit	All parameters	Parameters	Basic	Incorrect	20%
Configuration	rate, sequence length,	are accurately set	are mostly	parameters	parameter	
& Execution	bits per symbol, laser	for every format	accurate with	are	settings or	
	frequency (193.1 THz)	and all	minor	configured,	failure to run	
	and power, fiber	simulations are	discrepancies	but several	the	
	length, carrier	executed	(e.g., sequence	settings are	simulations.	
	frequency and filter	successfully.	length or bit	missing or		
	and threshold-detector		rate).	incorrect.		
	settings.					
Measurement	Accurate recording of	Precise data	Data is	Partial data	Inaccurate,	20%
& Waveform	Q-factor, BER, eye	extraction for all	collected with	collection with	incomplete or	
Monitoring	height and threshold	formats, neatly	minor	incomplete	missing data	
	for OOK, and	tabulated with	omissions in	tables or	and	
	observation of M-ary	clear waveforms,	readings or	missing	measurements.	
	signal levels, eye	eye diagrams	visualizer	diagrams.		
	diagrams, constellation	and constellation	captures.			
	diagrams and spectra	diagrams.				
	for PAM-4, DPSK and					
	QAM.					
Modulation	Evaluation of bits per	Comprehensive	Clear analysis	Basic	Erroneous or	20%
Performance	symbol, symbol rate,	analysis correctly	identifying	comparison	missing	

Analysis	spectral efficiency, eye	relating bits per	basic trends	without	analysis of	
	opening and	symbol to	among the	explaining the	simulation	
	constellation spacing,	spectral	formats.	trade-off	results.	
	and their effect on	efficiency, eye		between		
	noise tolerance for each	opening and		spectral		
	format.	noise tolerance		efficiency and		
		with sound		noise		
		explanations.		tolerance.		
Report &	Comparison of the four	Report features	Satisfactory	Incomplete	Disorganized	20%
Conclusion	formats, identification	clear diagrams,	report with	report with	or incomplete	
	of a suitable format for	complete tables	relevant	missing	report lacking	
	a given application,	and insightful	displays and a	graphs,	required	
	and structural clarity.	conclusions on	sound	readings or	diagrams and	
		the choice of	concluding	weak	valid	
		modulation	summary.	conclusions.	conclusions.	
		format.				

Result: The advanced optical modulation formats were successfully simulated and analyzed using OptiSystem. The performance of different modulation formats was observed and compared based on signal quality and transmission performance.